BEFORE THE INDIAN PATENT OFFICE

Under the Patents Act, 1970 and the Patents Rules, 2003

(As amended up to date)

PROVISIONAL SPECIFICATION

(See Section 9 and Rule 18)

TITLE OF INVENTION:

AN INTEGRATED AGRICULTURAL INTELLIGENCE AND INVESTMENT PLATFORM COMPRISING AN ARTIFICIAL-INTELLIGENCE-DRIVEN AGRICULTURAL PORTFOLIO OPTIMISATION SYSTEM, A SATELLITE-VALIDATED MULTI-STAGE LAND TOKENISATION ENGINE, A ROLE-ISOLATED DUAL-APPLICATION SHARED-BACKEND ARCHITECTURE, AND A CLOSED-LOOP FARM-HEALTH-SIGNAL DRIVEN PORTFOLIO REBALANCING METHOD

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This document is a **Provisional Patent Application** filed under Section 9 of the Patents Act, 1970. It establishes the priority date for the inventions described herein. A Complete Specification must be filed within **12 months** of the provisional filing date under Section 9(1). All information contained herein is strictly confidential. Unauthorised disclosure may constitute a breach of trade secret and intellectual property law.

Version 1.0 | June 2026 | Aditi Gaikwad & Pratik Jadhav

Contents

1. TITLE OF THE INVENTION

AN INTEGRATED AGRICULTURAL INTELLIGENCE AND INVESTMENT PLATFORM COMPRISING AN ARTIFICIAL-INTELLIGENCE-DRIVEN AGRICULTURAL PORTFOLIO OPTIMISATION SYSTEM, A SATELLITE-VALIDATED MULTI-STAGE LAND TOKENISATION ENGINE, A ROLE-ISOLATED DUAL-APPLICATION SHARED-BACKEND ARCHITECTURE, AND A CLOSED-LOOP FARM-HEALTH-SIGNAL-DRIVEN PORTFOLIO REBALANCING METHOD.

2. FIELD OF THE INVENTION

The present invention relates to the field of agricultural technology, agricultural finance, and precision farming intelligence systems. More specifically, the invention relates to:

(i) a system and method for constructing diversified agricultural investment portfolios using multi-dimensional, satellite-validated farm-level intelligence data processed through a constrained optimisation engine;

(ii) a system and method for digitising agricultural land assets into fractional ownership units through a multi-stage verification pipeline incorporating geofence spatial matching against government cadastral databases and satellite-derived vegetation index validation as mandatory eligibility gates;

(iii) a unified agricultural intelligence platform architecture wherein two independent role-isolated mobile applications — a farmer application and an investor application — share a common artificial intelligence and market intelligence backend with role-partitioned data access controls; and

(iv) a method for automatically triggering portfolio rebalancing decisions in an agricultural investment system based on satellite-derived farm health signals and composite farm health scores computed from live remote sensing, soil, sensor, and risk data.

3. BACKGROUND OF THE INVENTION

3.1. Agricultural Information Asymmetry

Farmers in India and developing agricultural economies make high-stakes operational decisions — crop selection, irrigation scheduling, fertiliser application, and harvest timing — using fragmented, stale, or absent data. Soil reports, weather forecasts, commodity market prices, and government advisories exist in disconnected silos in incompatible formats. By the time actionable intelligence reaches a farmer, the optimal intervention window has typically passed. No existing system aggregates satellite remote sensing, IoT sensor telemetry, climate forecasts, commodity market signals, and government agricultural databases into a unified, personalised, farm-level intelligence layer delivered through mobile.

3.2. Agricultural Investment Opacity

Institutional and retail investors seeking exposure to agricultural assets face fundamental structural barriers:

- **No standardised farm score:** No data-driven, satellite-validated scoring system exists to evaluate farm-level health, resilience, or return potential at a granularity suitable for investment decisions.
- **No portfolio construction tools:** Agricultural investment lacks the portfolio diversification, risk management, and rebalancing mechanisms available in listed equity markets. Existing tools treat agricultural exposure as a single asset class rather than a universe of individually scoreable positions.
- **Illiquidity and opacity:** Agricultural assets are illiquid, non-fungible, geographically dispersed, and lack secondary markets for exit.
- **No fractional ownership:** Retail investors cannot participate in agricultural assets at sub-land-acquisition ticket sizes.
- **No satellite-validated verification:** No prior system uses remote sensing vegetation indices as a mandatory eligibility gate in the tokenisation of agricultural assets.

3.3. Deficiencies of Prior Art

Existing agricultural technology platforms address isolated sub-problems: crop advisory applications provide recommendations without satellite validation or market intelligence integration; commodity price aggregators provide market data without farm-level context; land record digitisation projects create digital records without investment instruments or fractional ownership infrastructure; general-purpose portfolio optimisation systems address financial assets and do not incorporate agricultural-specific diversity dimensions such as crop type, harvest month, water source dependency, or disease exposure. No prior art discloses a unified system that: (a) integrates satellite remote sensing, IoT sensors, climate data, and commodity markets into a shared intelligence backend serving both a farmer and an investor application; (b) uses satellite-derived vegetation indices as a mandatory eligibility gate in a land tokenisation workflow; (c) constructs agricultural investment portfolios optimised across crop, geographic, climate, harvest-season, water, and disease diversity dimensions simultaneously; or (d) uses a composite farm health score derived from live satellite, soil, sensor, and risk data to automatically trigger investor portfolio rebalancing.

4. OBJECTS OF THE INVENTION

The principal objects of the present invention are:

O1. To provide an agricultural portfolio optimisation system that treats farm-level investment positions as a universe of individually scored assets and constructs diversified portfolios optimised simultaneously across crop type, geographic region, climate zone, harvest season, water dependency, and disease exposure dimensions.

O2. To provide a land tokenisation engine that enforces satellite-validated agricultural use verification — using Normalised Difference Vegetation Index (NDVI) computed from multi-spectral satellite imagery over a trailing twelve-month period — as a mandatory pre-tokenisation eligibility gate, in combination with government land record validation and geofence spatial matching against cadastral databases.

O3. To provide a unified architectural platform wherein a farmer intelligence application and an investor application share a common artificial intelligence engine, satellite engine, risk engine, and market intelligence engine, with role-partitioned access controls ensuring that investor-facing endpoints are invisible to farmer authentication tokens and vice versa.

O4. To provide a closed-loop method wherein a composite Farm Health Score — computed from live satellite-derived vegetation indices, soil data, IoT sensor readings, and multi-dimensional risk scores — automatically triggers portfolio rebalancing recommendations in the investor application when the score crosses defined thresholds.

O5. To provide a multi-stage stakeholder verification pipeline combining government API validation, biometric matching, geospatial polygon matching, and satellite confirmation into a single modular, composable verification system applicable to farmers, investors, and asset owners.

5. STATEMENT OF THE INVENTION

According to the present invention there is provided an integrated agricultural intelligence and investment platform, and associated systems and methods as described in the following sections and as claimed in the appended provisional claims.

The invention comprises four inter-related technical contributions, each independently patentable and collectively forming a unified platform:

Invention I

AI-Driven Agricultural Portfolio Optimisation System — a system for constructing and rebalancing diversified agricultural investment portfolios using constrained mean-variance optimisation augmented with a novel agricultural diversity bonus function.

Invention II

Satellite-Validated Land Tokenisation Engine — a multi-phase system for digitising agricultural land assets into fractional ownership tokens, enforcing NDVI-based agricultural use verification and geofence cadastral matching as mandatory eligibility gates.

Invention III

Role-Isolated Dual-Application Shared-Backend Architecture — a platform architecture wherein two independent mobile applications serving different user personas share a unified artificial intelligence backend with attribute-based, role-isolated access controls.

Invention IV

Closed-Loop Farm-Health-Signal-Driven Portfolio Rebalancing — a method wherein satellite-derived and sensor-derived composite Farm Health Scores propagate automatically from the farmer intelligence layer to the investor portfolio layer, triggering rebalancing actions without human intervention.

6. DETAILED DESCRIPTION OF THE INVENTION

6.1. Invention I: AI-Driven Agricultural Portfolio Optimisation System

6.1.1. Overview

The AI Portfolio Engine constructs diversified agricultural investment portfolios analogous to mutual funds or exchange-traded funds (ETFs), where the underlying assets are farm-level investment positions rather than listed securities. The system accepts investor preference inputs, constructs a universe of eligible farm positions, scores each position across multiple dimensions, computes a pairwise diversity matrix, and solves a constrained optimisation problem to produce an optimal portfolio specification.

6.1.2. Investor Input Parameters

The system accepts the following investor inputs: investment amount (in Indian Rupees); risk appetite on a scale of one to ten; preferred Indian states for investment; preferred crop categories; investment horizon in months; liquidity preference represented as lock-in tolerance; and environmental, social, and governance (ESG) preference score in the range $[0, 1]$.

6.1.3. Universe Construction

From the complete set of registered and verified farm positions, the system constructs an eligible investment universe by applying the following filters simultaneously:

(a) Verification status must be `APPROVED` as determined by the multi-stage verification pipeline;

(b) Investor Score $S_{\text{investor}} > 0.50$;

(c) State preference match against investor-specified regions;

(d) Crop category preference match.

6.1.4. Feature Extraction

For each farm position in the eligible universe, the system extracts the following feature vector:

$$\mathbf{f}_i = \left[Y_i,\ R_i,\ C_i,\ H_i,\ M_i,\ G_i,\ W_i,\ D_i,\ I_i,\ \sigma_i^2,\ t_i^{\text{harvest}}\right] \tag{1}$$

where Y_i is yield confidence score, R_i is risk score, C_i is climate resilience score, H_i is satellite health trend score, M_i is market demand score, G_i is carbon sequestration score, W_i is water dependency index, D_i is disease exposure index, I_i is insurance coverage indicator, σ_i^2 is historical yield variance, and t_i^{harvest} is expected harvest month.

6.1.5. Agricultural Diversity Scoring

A pairwise diversity matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is computed across all candidate farm pairs on the following agricultural dimensions, which constitute a novel contribution of the present invention:

(i) **Crop-type diversity:** positions in different crop families receive higher diversity bonus;

(ii) **Geographic region diversity:** positions in different districts and agro-climatic zones;

(iii) **Climate zone diversity:** positions across different Köppen climate classifications;

(iv) **Harvest month diversity:** positions with non-overlapping harvest windows, reducing cashflow concentration risk;

(v) **Water source diversity:** positions using different irrigation sources (rainfed, canal, groundwater, drip);

(vi) **Disease profile diversity:** positions with low pairwise disease co-exposure, reducing correlated loss risk.

The diversity bonus function is defined as:

$$D(\mathbf{w}) = \mathbf{w}^\top \mathbf{A}\, \mathbf{w} \tag{2}$$

where $\mathbf{w} \in \mathbb{R}^n$ is the portfolio weight vector.

6.1.6. Portfolio Optimisation

The system solves the following constrained optimisation problem:

$$\max_{\mathbf{w}} \quad \mathbf{w}^\top \boldsymbol{\mu} - \lambda\, \mathbf{w}^\top \boldsymbol{\Sigma}\, \mathbf{w} + \gamma\, D(\mathbf{w}) \tag{3}$$

subject to:

$$\sum_{i=1}^{n} w_i = 1 \tag{4}$$

$$w_i \geq 0 \quad \forall i \tag{5}$$

$$w_i \leq w_{\max} \quad \forall i \tag{6}$$

where $\boldsymbol{\mu} \in \mathbb{R}^n$ is the expected return vector for each farm position derived from the Yield Engine and Market Intelligence Engine; $\boldsymbol{\Sigma} \in \mathbb{R}^{n \times n}$ is the risk covariance matrix computed from historical yield variance and pairwise climate and market correlations; $\lambda > 0$ is the risk aversion parameter mapped monotonically from the investor's risk appetite input; $\gamma > 0$ is the diversity weight; $w_{\max}$ is the maximum concentration limit per farm position; and additional constraints enforce liquidity and horizon requirements. The parameter γ is a novel element of this optimisation formulation, specifically designed for agricultural portfolio diversification across the six dimensions described in §6.1.5.

6.1.7. Portfolio Templates

The system provides five pre-defined portfolio templates that override λ and γ with preset values and apply additional agricultural-domain constraints:

Template	λ	γ	**Additional Constraint**
Conservative	High	Medium	$R_i < 0.40$; insurance required
Balanced	Medium	Medium	No additional
Growth	Low	Medium	$Y_i > 0.65$
ESG-Focused	Medium	High	$G_i > 0.70$; no chemical-intensive crops
High-Yield	Low	Low	$M_i > 0.70$; $Y_i > 0.70$

6.1.8. Investor Score Formula

The Investor Score assigned to each farm position for portfolio eligibility is computed as:

$$S_{\text{investor}} = \mathbf{v}^{\top} \Big[S_{\text{health}},\ 1{-}S_{\text{risk}},\ S_{\text{yield}},\ S_{\text{sat}},\ S_{\text{clim}},\ S_{\text{carbon}},\ S_{\text{opp}} \Big]^{\top} \tag{7}$$

where $\mathbf{v} = [0.20, 0.20, 0.15, 0.15, 0.10, 0.10, 0.10]$ are the weighting coefficients.

6.2. Invention II: Satellite-Validated Land Tokenisation Engine

6.2.1. Overview

The Land Tokenisation Engine digitises approved agricultural land assets into fractional digital ownership units through a six-phase lifecycle. The novel technical contribution of this invention is the mandatory use of satellite-derived Normalised Difference Vegetation Index (NDVI) as an eligibility gate in Phase 1, in combination with geofence polygon spatial matching against government cadastral databases, prior to any tokenisation action.

6.2.2. Phase 1: Multi-Stage Asset Verification

Asset verification proceeds through the following sequential stages, each implemented as an independent microservice returning a pass/fail/pending result with a confidence score:

V1. Government Land Record Validation: Cross-reference uploaded land record data (7/12 extract, patta, khatauni) against state revenue authority APIs.

V2. Survey Number Validation: Cross-check survey number against the national cadastral database.

V3. Geofence Polygon Spatial Matching: The asset owner uploads a polygon boundary (GeoJSON format). The system computes the spatial intersection of the submitted polygon against the government survey polygon retrieved from the cadastral database using PostGIS `ST_Intersection`. A match threshold of $\geq$ 85% spatial overlap is enforced. Polygons with $<$ 85% overlap are automatically rejected, preventing fraudulent asset boundary manipulation.

V4. Ownership Verification: Title search, encumbrance check, and lien check via state revenue APIs.

V5. KYC of Asset Owner: Identity verification of the individual or entity registering the asset.

V6. AML Screening: Sanctions list and Politically Exposed Person (PEP) check.

V7. Legal Due Diligence: Third-party legal opinion on clear and marketable title.

V8. Satellite Vegetation Validation (Novel Eligibility Gate): The system retrieves multi-spectral satellite imagery (Sentinel-2 or equivalent) for the submitted farm polygon over the trailing twelve-month period. For each cloud-free scene, the system computes the NDVI at pixel level as:

$$\text{NDVI} = \frac{\rho_{\text{NIR}} - \rho_{\text{RED}}}{\rho_{\text{NIR}} + \rho_{\text{RED}}} \tag{8}$$

where ρ_{NIR} is near-infrared band reflectance and ρ_{RED} is red band reflectance. The polygon- averaged NDVI is computed for each scene. An asset is eligible for tokenisation *if and only if* the polygon-averaged NDVI exceeds 0.3 for a statistically significant proportion of cloud-free scenes over the trailing twelve-month window, confirming active agricultural use. Assets with NDVI consistently below this threshold — indicating bare land, non-agricultural use, or deception — are automatically rejected.

V9. Optional Physical Inspection: On-ground verification by a platform-empanelled inspector.

6.2.3. Phase 2: Fractional Token Creation

Upon successful completion of Phase 1 verification, the system:

(a) Registers the asset in a token registry database with a unique `asset_id`;

(b) Denominates total ownership into N fractional units (configurable; default $N = 10{,}000$ per hectare);

(c) Assigns each unit a unique `token_id` linked to `asset_id`;

(d) Generates a Digital Ownership Certificate (PDF with QR code) linked to the immutable ownership ledger;

(e) Creates an immutable ownership ledger entry recording the creation event with timestamp, verifier identity, and cryptographic evidence hashes.

6.2.4. Phases 3–6: Marketplace Listing, Trading, Settlement, and Token Burn

Phases 3 through 6 cover: marketplace listing with satellite thumbnail and verification badge; primary sale, secondary market trading, and peer-to-peer transfer through an escrow-based settlement system with T+1 settlement and immutable ownership chain; and exit via pro-rata proceeds distribution and token burn (marking all tokens `BURNED` in the registry, preventing further trading).

6.2.5. Fraud Detection Subsystem

The tokenisation engine incorporates an automated fraud detection subsystem comprising:

(i) **Duplicate asset detection:** Polygon overlap check against all registered assets; overlap $> 20\%$ triggers automatic flag.

(ii) **Ownership chain validation:** Verification of unbroken ownership chain from creation event.

(iii) **Velocity checks:** Flagging of tokens transferred more than three times within a twenty-four hour window.

(iv) **Price anomaly detection:** Flagging of trade prices deviating more than 50% from estimated fair value.

6.3. Invention III: Role-Isolated Dual-Application Shared-Backend Architecture

6.3.1. Overview

The platform architecture provides two independent mobile applications — a farmer application (*UniAgriQ*) and an investor application (*Genesis*) — that share a unified backend comprising artificial intelligence engines, satellite analytics, market intelligence, risk computation, authentication infrastructure, and polyglot database storage, while enforcing strict role-based isolation at the API gateway layer such that each application's endpoints are architecturally invisible to the authentication tokens of the other application.

6.3.2. Shared Intelligence Backend

The shared backend comprises the following engines, each implemented as an independent, horizontally scalable microservice:

- **AI/ML Engine:** Model inference service, feature store, and model registry. Serves crop recommendation (farmer application) and portfolio optimisation (investor application) from the same model infrastructure.

- **Market Intelligence Engine:** Commodity price trend analysis, demand forecasting, and opportunity alerting. Consumed by farmer sell-timing recommendations and investor return projections simultaneously.
- **Satellite Engine:** NDVI, Enhanced Vegetation Index (EVI), and Normalised Difference Moisture Index (NDMI) computation at farm-polygon resolution. Used for farmer health dashboards and investor asset validation.
- **Risk Engine:** Multi-dimensional risk scoring across eight dimensions. Powers farmer risk dashboard and investor portfolio risk assessment from the same underlying computation.
- **Notification Service:** Push, SMS, in-app, and email notification infrastructure with role-specific routing rules.

6.3.3. Role-Isolated Access Control

The API Gateway enforces the following role isolation model:

(a) Farmer authentication tokens (`role: FARMER`) are permitted access only to farmer-designated endpoints; investor endpoints return `HTTP 403` for farmer tokens.

(b) Investor authentication tokens (`role: INVESTOR`) are permitted access only to investor-designated endpoints; farmer endpoints return `HTTP 403` for investor tokens.

(c) A hybrid Role-Based Access Control (RBAC) and Attribute-Based Access Control (ABAC) model is applied: RBAC defines baseline permissions per role; ABAC policies add fine-grained constraints (e.g., an investor token can access farm data *only* for farms within their active portfolio positions).

(d) Mutual TLS (mTLS) is enforced between all internal microservices; no implicit trust is assumed between services.

6.3.4. Twelve-Layer Reference Architecture

The platform is organised into twelve architectural layers:

L1. Presentation: UniAgriQ App + Genesis App (independent mobile applications)

L2. Application: API Gateway, Backend-for-Frontend (BFF), Orchestration

L3. AI/ML: Inference, Feature Store, Model Registry

L4. Intelligence: Crop, Market, Risk, Yield, Carbon, Climate engines

L5. Investment: Portfolio, Tokenisation, Collective Investment, Marketplace engines

L6. Data Ingestion: ETL pipelines, stream processing, normalisation

L7. External Data: APIs, satellite providers, IoT, market feeds

L8. Storage: Relational, time-series, document, vector, object storage

L9. Notification: Push, SMS, in-app, email

L10. Security: Authentication, authorisation, KMS, fraud detection, audit

L11. Financial: Escrow, settlement, wallet, transaction ledger

L12. Observability: Logging, metrics, distributed tracing, alerting

6.4. Invention IV: Closed-Loop Farm-Health-Signal-Driven Portfolio Rebalancing

6.4.1. Overview

The present invention provides a method wherein a composite Farm Health Score — computed from live satellite-derived vegetation indices, soil data, IoT sensor readings, and multi-dimensional risk scores — is continuously monitored and, upon crossing defined thresholds, automatically triggers portfolio rebalancing recommendations in the investor application, without requiring any manual human intervention.

6.4.2. Composite Farm Health Score Computation

The Farm Health Score S_{health} for a given farm at time t is computed as:

$$S_{\text{health}}(t) = 0.25\,\phi(\Delta\text{NDVI}(t)) + 0.20\,\phi(S_{\text{soil}}) + 0.15\,\phi(S_{\text{sensor}}(t)) + 0.25\left(1 - \phi(S_{\text{risk}}(t))\right) + 0.15\,\phi(S_{\text{cal}}(t) \tag{9}$$

where:

- $\phi(\Delta\text{NDVI}(t))$ is the normalised change in polygon-averaged NDVI relative to the twelve-month baseline, derived from live satellite imagery;

- $\phi(S_{\text{soil}})$ is the normalised soil health sub-score derived from the most recent soil report;
- $\phi(S_{\text{sensor}}(t))$ is the normalised IoT sensor health sub-score aggregated from real-time soil moisture, temperature, and humidity readings;
- $\phi(S_{\text{risk}}(t))$ is the normalised composite risk score from the Risk Engine, inverted so that lower risk contributes positively to health; and
- $\phi(S_{\text{cal}}(t))$ is the calendar adherence score measuring deviation from the recommended crop calendar.

The normalisation function $\phi : \mathbb{R} \to [0, 1]$ maps each raw sub-score to a unit interval.

6.4.3. Rebalancing Trigger Conditions

The system monitors $S_{\text{health}}(t)$ for each farm position in every active investor portfolio at a cadence of no less than once per hour for real-time signals (satellite anomalies, sensor alerts) and daily for computed scores. A portfolio rebalancing action is automatically triggered when any of the following conditions are met for a farm position i in portfolio p:

T1. Health Score Degradation: $S_{\text{health},i}(t)$ falls below $S_{\text{health},i}(t - 30\text{d}) - 0.20$, i.e., a decline of more than 0.20 units over a thirty-day rolling window.

T2. Risk Score Spike: The per-dimension risk score for farm i increases by more than 20% relative to its thirty-day moving average.

T3. Satellite Anomaly Detection: The satellite engine detects an NDVI z-score of $|z| > 2.0$ relative to the ten-year historical baseline for farm i's polygon.

T4. Harvest Completion: Farm i completes its harvest cycle, releasing its weight for reallocation.

T5. New High-Opportunity Entry: A new farm position enters the eligible universe with $S_{\text{investor}} > 0.75$ and a projected contribution that improves the portfolio's Sharpe ratio.

6.4.4. Rebalancing Execution

Upon trigger, the system re-solves the portfolio optimisation problem defined in eq. (3) with the updated feature vectors, flags the degraded or exited position for weight reduction, and presents the investor with:

(a) the specific trigger condition that initiated rebalancing;

(b) the recommended new portfolio weight vector $\mathbf{w}^*_{\text{new}}$;

(c) the projected impact on expected return, risk, and diversification score; and

(d) actionable buttons within the investor application to accept, modify, or reject the rebalancing recommendation.

The closed-loop nature of this system — wherein live satellite and sensor signals from the farmer intelligence layer propagate directly to portfolio rebalancing decisions in the investor layer through the shared backend — constitutes a novel technical contribution not present in any prior art.

7. BRIEF DESCRIPTION OF THE DRAWINGS

The following figures, to be included in the Complete Specification, illustrate the present invention. Reference numbers are consistent across all figures.

Figure 1
Dual-Application, Shared-Backend Platform Architecture showing the UniAgriQ Farmer Application and Genesis Investor Application connected to the unified backend through the API Gateway.

Figure 2
AI Portfolio Engine Construction and Rebalancing Pipeline showing the six-step flow from investor inputs through universe construction, feature extraction, diversity scoring, portfolio optimisation, template matching, and rebalancing monitor.

Figure 3
Land Tokenisation Lifecycle showing all six phases from asset verification through token creation, marketplace listing, trading, settlement, and token burn.

Figure 4
Satellite Vegetation Validation Workflow showing the NDVI computation pipeline from satellite imagery ingestion to agricultural use eligibility determination.

Figure 5
Closed-Loop Farm-Health-Signal Rebalancing Method showing the data flow

from satellite/sensor signals through Farm Health Score computation to portfolio rebalancing trigger and execution.

Figure 6
Role-Isolated API Gateway Architecture showing role-based endpoint visibility enforcement between farmer and investor tokens.

Figure 7
Twelve-Layer Reference Architecture of the Unified Platform.

Figure 8
Multi-Stage Farmer Verification Pipeline.

Figure 9
Multi-Stage Investor Verification Pipeline.

Figure 10
Escrow-Based Settlement Flow for Token Trades.

8. PROVISIONAL CLAIMS

Drafting Note: The following provisional claims are intentionally broad to maximise priority date coverage. Narrower dependent claims and method claims will be fully elaborated in the Complete Specification filed within twelve months. Claims are framed as systems and methods producing technical effects on physical agricultural assets and financial instruments to comply with Section 3(k) of the Patents Act, 1970 and the CRI Guidelines, 2017.

CLAIMS:

1. *[Invention I — Independent System Claim]*
An agricultural investment portfolio construction system comprising:

(a) a universe construction module configured to filter, from a database of verified farm positions, an eligible investment universe based on verification status, investor score threshold, geographic preference, and crop preference parameters supplied by an investor;

(b) a feature extraction module configured to extract, for each farm position in said eligible universe, a multi-dimensional feature vector comprising yield confidence score, risk score, climate resilience score, satellite-derived health trend score, market demand score, carbon sequestration score, water dependency index, disease exposure index, insurance coverage indicator, historical yield variance, and expected harvest month;

(c) a diversity scoring module configured to compute a pairwise agricultural diversity matrix across said farm positions on at least the dimensions of crop type, geographic region, climate zone, harvest month, water source, and disease profile;

(d) a portfolio optimisation engine configured to solve a constrained optimisation problem to determine portfolio weight allocations across said farm positions, wherein the objective function maximises expected return less a risk-aversion-weighted variance term plus a diversity bonus term computed from said agricultural diversity matrix; and

(e) an output module configured to generate a portfolio specification comprising per-farm allocation weights, expected return, risk contribution, harvest timeline, and diversification metrics.

2. *[Invention I — Dependent Claim]*
The system of claim 1, wherein the objective function is:

$$\max_{\mathbf{w}} \quad \mathbf{w}^\top \boldsymbol{\mu} - \lambda\, \mathbf{w}^\top \boldsymbol{\Sigma}\, \mathbf{w} + \gamma\, \mathbf{w}^\top \mathbf{A}\, \mathbf{w}$$

subject to $\sum_i w_i = 1$, $w_i \geq 0$, and $w_i \leq w_{\max}$ for all i, wherein λ is mapped from investor risk appetite, γ controls the weight of the agricultural diversity bonus, $\boldsymbol{\Sigma}$ is a risk covariance matrix, $\boldsymbol{\mu}$ is the expected return vector, and $\mathbf{A}$ is the pairwise agricultural diversity matrix.

3. *[Invention I — Dependent Claim: Templates]*
The system of claim 1, further comprising a template matching module providing pre-defined portfolio templates including at minimum a conservative template, a balanced template, a growth template, an ESG-focused template, and a high-yield template, each overriding the risk aversion and diversity parameters with preset values and applying additional agricultural domain-specific constraints.

4. *[Invention I — Independent Method Claim]*
A computer-implemented method for constructing a diversified agricultural investment portfolio, the method comprising:

(a) receiving investor preference inputs comprising investment amount, risk appetite, geographic preferences, crop preferences, investment horizon, liquidity preference, and ESG preference;

(b) filtering a database of verified farm positions to construct an eligible investment universe;

(c) extracting a multi-dimensional feature vector for each eligible farm position from satellite imagery, IoT sensor data, soil reports, market intelligence data, and yield prediction data;

(d) computing an agricultural diversity matrix across said farm positions on at least six agricultural dimensions comprising crop type, geographic region, climate zone, harvest month, water source, and disease profile;

(e) solving a constrained optimisation problem incorporating said agricultural diversity matrix to produce portfolio weight allocations; and

(f) outputting a portfolio specification and presenting same to the investor.

5. *[Invention II — Independent System Claim]*
A land tokenisation system for agricultural assets, comprising:

(a) a geofence polygon matching module configured to receive a GeoJSON polygon boundary submitted by an asset owner, retrieve the corresponding government survey polygon from a cadastral database, compute the spatial intersection of said polygons, and reject the asset if the spatial overlap falls below a threshold of eighty-five percent;

(b) a satellite vegetation validation module configured to retrieve multi-spectral satellite imagery for the submitted polygon over a trailing twelve-month period, compute the Normalised Difference Vegetation Index (NDVI) for each cloud-free scene, compute a polygon-averaged NDVI, and reject the asset from tokenisation eligibility if said polygon-averaged NDVI does not exceed a threshold of 0.3 over said period;

(c) a verification orchestrator configured to enforce completion of government land record validation, survey number cross-check, geofence polygon matching, ownership verification, KYC verification, AML screening, and legal due diligence prior to any tokenisation action; and

(d) a token creation module configured, upon successful completion of all verification stages, to denominate the verified asset into fractional ownership units, assign unique token identifiers, generate digital ownership certificates, and create immutable ledger entries recording the creation event.

6. *[Invention II — Dependent Claim: Fraud Detection]*
The system of claim 5, further comprising a fraud detection module configured to: detect duplicate asset registration by checking polygon overlap against all registered assets and flagging overlap exceeding twenty percent; validate unbroken ownership chain from creation event; flag tokens transferred more than three times within a twenty-four hour period; and flag trade prices deviating more than fifty percent from estimated fair value.

7. *[Invention II — Dependent Claim: Escrow Settlement]*
The system of claim 5, further comprising an escrow-based settlement module configured to hold buyer funds in escrow upon order placement, transfer ownership in the token registry upon settlement confirmation, release funds to the seller upon successful ownership transfer, record each state change in an append-only immutable transaction ledger with cryptographic hash linking to the prior entry, and automatically refund escrowed funds if settlement is not completed within a configurable timeout period.

8. *[Invention II — Independent Method Claim]*
A computer-implemented method for tokenising agricultural land assets, the method comprising:

(a) receiving a polygon boundary from an asset owner and computing spatial intersection against a government cadastral survey polygon, rejecting the asset if overlap is below eighty-five percent;

(b) retrieving multi-spectral satellite imagery for said polygon over a trailing twelve-month period and computing a polygon-averaged Normalised Difference Vegetation Index, rejecting the asset if said index does not exceed 0.3;

(c) completing government land record validation, ownership verification, KYC, AML screening, and legal due diligence;

(d) upon successful completion of all verification stages, denominating the asset into fractional digital ownership units with unique identifiers;

(e) creating an immutable ledger entry recording the tokenisation event; and

(f) listing the fractional units on a marketplace for primary sale and subsequent secondary trading.

9. *[Invention III — Independent System Claim]*
An agricultural intelligence platform comprising:

(a) a first mobile application configured for farmer users, providing farm health dashboards, crop recommendations, climate intelligence, market intelligence, crop calendar generation, and conversational AI advisory;

(b) a second mobile application configured for investor users, providing agricultural portfolio construction, land token trading, collective investment management, marketplace access, and portfolio performance analytics;

(c) a unified backend comprising at least an artificial intelligence engine, a satellite engine, a market intelligence engine, a risk engine, an authentication service, a notification service, and a polyglot database layer, all shared between said first and second applications; and

(d) an API gateway configured to enforce role-isolated access control wherein authentication tokens issued to farmer users are denied access to investor-designated endpoints and authentication tokens issued to investor users are denied access to farmer-designated endpoints, such that each application's endpoint namespace is architecturally invisible to the other application's authentication context.

10. *[Invention III — Dependent Claim: Hybrid RBAC+ABAC]*
The platform of claim 9, wherein the API gateway enforces a hybrid access control model comprising role-based access control defining baseline permissions per user role, combined with attribute-based access control policies applying fine-grained constraints including a constraint that an investor user may access farm-level data only for farms within said investor's active portfolio positions.

11. *[Invention III — Dependent Claim: Shared Intelligence]*
The platform of claim 9, wherein the satellite engine computes vegetation indices for a given farm polygon and said computed indices are consumed simultaneously by the first application for farmer health dashboard presentation and by the second application for investor asset validation and portfolio risk assessment, without duplicating the underlying computation or storage.

12. *[Invention IV — Independent Method Claim]*
A computer-implemented method for automatically rebalancing an agricultural investment portfolio based on farm health signals, the method comprising:

(a) computing, for each farm position in an active investor portfolio, a composite Farm Health Score from at least: satellite-derived vegetation index change, soil health sub-score, IoT sensor health sub-score, multi-dimensional risk score, and crop calendar adherence score;

(b) continuously monitoring said composite Farm Health Score at a cadence of no less than daily;

(c) automatically triggering a portfolio rebalancing action when any of the following conditions are detected for a farm position: (i) the Farm Health Score declines by more than 0.20 units over a thirty-day rolling window; (ii) any per-dimension risk score increases by more than twenty percent relative to its thirty-day moving average; (iii) the satellite engine detects an NDVI z-score exceeding $|z| > 2.0$ relative to a ten-year historical baseline; (iv) the farm completes its harvest cycle; or (v) a new farm position enters the eligible universe with a projected contribution improving the portfolio Sharpe ratio;

(d) re-solving the portfolio optimisation problem with updated feature vectors and constraint parameters; and

(e) presenting the investor with the trigger condition, the recommended new portfolio weight vector, and the projected impact on expected return, risk, and diversification score.

13. *[Invention IV — Dependent Claim: Health Score Formula]*
The method of claim 12, wherein the composite Farm Health Score is computed as:

$$S_{\text{health}}(t) = w_1\,\phi(\Delta\text{NDVI}(t)) + w_2\,\phi(S_{\text{soil}}) + w_3\,\phi(S_{\text{sensor}}(t)) + w_4\left(1 - \phi(S_{\text{risk}}(t))\right) + w_5\,\phi(S_{\text{cal}}(t))$$

where $\phi : \mathbb{R} \to [0,1]$ is a normalisation function, $\Delta\text{NDVI}(t)$ is the change in polygon-averaged NDVI relative to a twelve-month baseline, S_{soil} is derived from the most recent soil report, $S_{\text{sensor}}(t)$ is aggregated from real-time IoT readings, $S_{\text{risk}}(t)$ is the multi-dimensional risk score, $S_{\text{cal}}(t)$ is the crop calendar adherence score, and $\sum_{j=1}^{5} w_j = 1$.

14. *[Invention IV — Dependent Claim: Closed-Loop Propagation]*
The method of claim 12, wherein said satellite-derived vegetation indices and said IoT sensor readings are generated by the same satellite engine and IoT processing module that serve the farmer application, such that farm health signals propagate from the farmer intelligence layer to the investor portfolio layer through a shared backend without requiring separate data collection, separate storage, or manual data transfer between applications.

15. *[Cross-Invention — System Claim]*
An integrated agricultural intelligence and investment platform comprising the agricultural portfolio construction system of claim 1, the land tokenisation system of claim 5, the dual-application platform of claim 9, and the portfolio rebalancing method of claim 12, operating as a unified system wherein:

(a) farm positions tokenised by the land tokenisation system are eligible for inclusion in portfolios constructed by the portfolio construction system;

(b) the portfolio rebalancing method monitors farm health scores for tokenised positions and triggers rebalancing when satellite or sensor signals indicate material change; and

(c) both the farmer application and the investor application consume intelligence from the shared backend with role-isolated access controls.

9. ABSTRACT OF THE INVENTION

An integrated agricultural intelligence and investment platform comprising: (I) an AI-driven agricultural portfolio optimisation system that constructs diversified farm investment portfolios by solving a constrained optimisation problem incorporating a novel agricultural diversity bonus function computed across crop type, geographic region, climate zone, harvest month, water source, and disease profile dimensions; (II) a satellite-validated land tokenisation engine that digitises agricultural land into fractional ownership units, enforcing polygon-averaged NDVI exceeding 0.3 over twelve months and geofence spatial overlap exceeding 85% against government cadastral databases as mandatory eligibility gates; (III) a role-isolated dual-application shared-backend architecture wherein a farmer application and an investor application share unified AI, satellite, market, and risk engines with attribute-based access controls ensuring endpoint invisibility between roles; and (IV) a closed-loop portfolio rebalancing method wherein composite Farm Health Scores computed from live satellite imagery, IoT sensors, soil data, and risk scores automatically trigger portfolio rebalancing when threshold conditions are met. The platform serves farmers through the UniAgriQ application and investors through the Genesis application.

10. DECLARATION AND UNDERTAKING

We, **Aditi Gaikwad** and **Pratik Jadhav**, Indian nationals, residing in Maharashtra, India, hereby declare:

(i) That we are the applicant(s) for this Provisional Patent Application.

(ii) That the invention described herein is our original invention and has not been published or communicated to any person or body prior to this filing, except as required for development and testing purposes under obligations of confidentiality.

(iii) That we undertake to file the Complete Specification within twelve (12) months from the date of filing of this Provisional Specification, in compliance with Section 9(1) of the Patents Act, 1970.

(iv) That we have not filed any application for a patent for the same or substantially the same invention in any other patent office, except as may be declared in Form 3.

(v) That we understand that this Provisional Specification establishes the priority date for the inventions described herein and does not by itself constitute a granted patent.

(vi) That the information provided in this application is true and correct to the best of our knowledge and belief.

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Name: Aditi Gaikwad

Designation: Inventor & Applicant

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Date: ______________________________

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Signed: ____________________________________

Date: ____________________________________

11. ANNEXURE: TECHNICAL GLOSSARY

For the purposes of this specification, the following terms are defined:

Term	**Definition**
NDVI	Normalised Difference Vegetation Index; a spectral index computed from near-infrared and red band reflectance values of satellite imagery; values range from -1 to $+1$; values > 0.3 indicate active vegetation.
EVI	Enhanced Vegetation Index; a spectral index that corrects for atmospheric and canopy background noise; more sensitive in high-biomass regions than NDVI.
NDMI	Normalised Difference Moisture Index; computed from near-infrared and shortwave infrared bands; indicates vegetation water content.
GeoJSON	An open standard format for encoding geographic data structures using JavaScript Object Notation.
PostGIS	A spatial database extension for PostgreSQL that adds support for geographic objects and spatial queries.
Cadastral Database	A government-maintained register of real property including survey numbers, boundaries, and ownership records.
Token	A digital representation of a fractional ownership unit in an agricultural land asset, recorded in an immutable ledger.
Token Burn	The permanent retirement of tokens, marking them as `BURNED` in the registry and preventing further trading.
Escrow	A financial arrangement wherein a third party holds funds on behalf of two transacting parties until settlement conditions are met.
CAP Table	A capitalisation table recording ownership percentages, contribution amounts, and voting rights for all investors in a collective investment vehicle.
Feature Store	A centralised repository for storing, managing, and serving machine learning features for both training and inference.
RAG	Retrieval-Augmented Generation; an AI architecture that retrieves relevant documents before generating a response, grounding the output in factual data.

RBAC	Role-Based Access Control; an access control method based on predefined user roles.
ABAC	Attribute-Based Access Control; an access control method based on attributes of users, resources, and environment conditions.
mTLS	Mutual Transport Layer Security; a protocol requiring both client and server to present certificates for authentication.
Sharpe Ratio	A measure of risk-adjusted return computed as the ratio of excess return to standard deviation of return.
Köppen Classification	A climate classification system that categorises climates based on temperature and precipitation patterns.
GDD	Growing Degree Days; a measure of heat accumulation used to predict plant development stages.
IoT	Internet of Things; a network of physical devices embedded with sensors and connectivity for data exchange.
KYC	Know Your Customer; a process of verifying the identity of clients in compliance with regulatory requirements.
AML	Anti-Money Laundering; regulations and procedures to prevent the generation of income through illegal actions.
PEP	Politically Exposed Person; an individual who holds or has held a prominent public function, subject to enhanced due diligence.

End of Provisional Specification

Aditi Gaikwad & Pratik Jadhav | June 2026